

Minimum Lap-Time Optimization of a Racing Line Using Optimal Control

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Problem Formulation

Objective

$$\min_{T, z(\cdot), u(\cdot)} T$$

States and controls

$$z = [s \quad e_y \quad e_\psi \quad v]^T$$

$$u = [a \quad \delta]^T$$

s : track progress,

e_y : lateral offset,

e_ψ : heading error,

v : velocity,

a : longitudinal acceleration,

δ : steering angle.

Vehicle dynamics

$$\dot{s} = \frac{v \cos(e_\psi)}{1 - \kappa e_y},$$

$$\dot{e}_y = v \sin(e_\psi),$$

$$\dot{e}_\psi = \frac{v}{L} \tan(\delta) - \kappa \dot{s},$$

$$\dot{v} = a.$$

Physical and Track Constraints

Track boundaries

$$-e_{y,\max} \leq e_y \leq e_{y,\max}$$

Velocity

$$0 \leq v \leq v_{\max}$$

Longitudinal acceleration

$$a_{\min} \leq a \leq a_{\max}$$

Steering

$$-\delta_{\max} \leq \delta \leq \delta_{\max}$$

Lateral acceleration

$$\left| \frac{v^2}{L} \tan(\delta) \right| \leq a_{y,\max}$$

Frenet validity

$$1 - \kappa e_y > 0$$

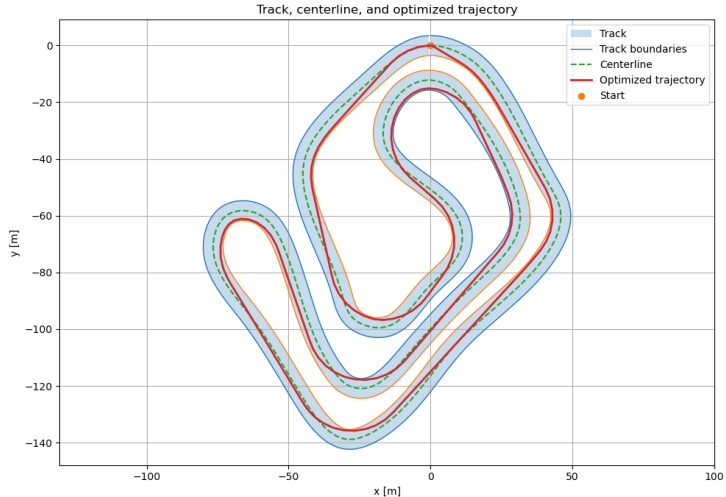
Numerical Solution

Direct multiple shooting

$$Z_{k+1} = F_{\text{RK4}}(Z_k, U_k, \Delta t), \quad \Delta t = \frac{T}{N}$$

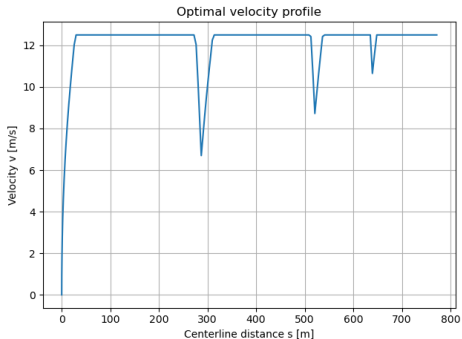
- The lap is divided into $N = 200$ shooting intervals.
- The states Z_k are optimized at every node.
- Vehicle dynamics are integrated with RK4.
- The resulting NLP is built in CasADi and solved with IPOPT.

Optimized Trajectory



The optimized path uses the available track width to increase the effective cornering radius.

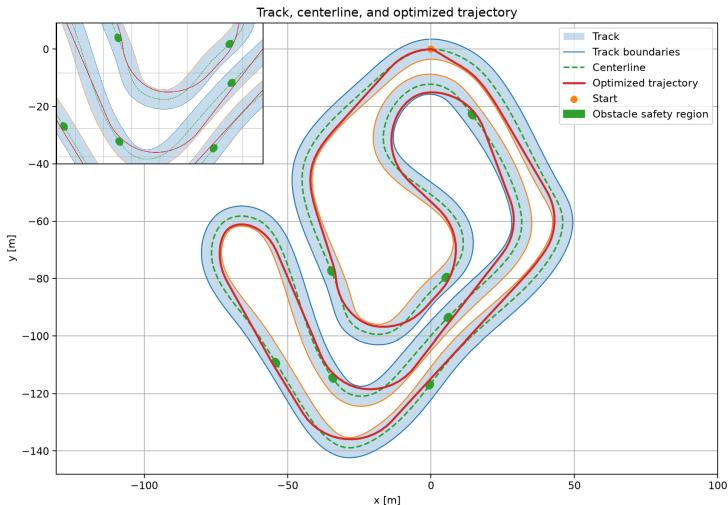
Results



Quantity	Value
Lap time	61.665 s
Maximum speed	12.5 m/s
Maximum offset	2.87 m
IPOPT iterations	357
Solver time	8.0 s

The kart reaches the speed limit on most straights and slows down in the high-curvature sections.

Extension: Static Obstacle Avoidance



The optimized trajectory deviates only locally to avoid the obstacles, increasing the lap time from 61.665 s to 61.800 s.

Conclusions

- The optimal control problem was successfully solved using direct multiple shooting.
- The optimized trajectory uses almost the full available track width.
- The kart reaches the maximum speed on the straights and slows down in the most demanding corners.
- The resulting minimum lap time is 61.665 s.
- The proposed formulation was successfully extended to include static obstacle avoidance.

Main result

Optimal control provides a feasible racing line that minimizes the lap time while satisfying the vehicle and track constraints.